(6) $x^2+y^2-4x-4y+8=0$ (7) $x^2+y^2-5x+8y+16=0$

(8) $y^2=2x$ (10) $8x^2+36y^2-16x+252y+431=0$

(11) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ (12) $9x^2-33x-3y+35=0$

(13) $x^2+y^2-12x+9y+63=0$, an imaginary circle

(14) $(5-\sqrt{21},3),(5+\sqrt{21},3),(8,-3),(2,-3)$ (15) $\frac{x^2}{25}+\frac{y^2}{9}=1$

Exercise 6.2

(1)(i) $y=5x-4$ (ii) $3x-y=2$ (iii) $2x+3y=5$ (iv) $x+\sqrt{3}y=(1+\sqrt{3})$

(3) $10x+3y=25$

(5)(i) $C=\frac{5}{9}(F-32)$ (or) $F=\frac{9}{5}C+32$ (ii) $C=37^\circ$ (iii) $F=100.4^\circ$

(6)(i) 4400metre (ii) $D=3000$metre (iii) $T=22seconds$ (7) $P=1,55,000$

(8) $\sqrt{(3)}x+y=24$ (9) $3x-4y=12, x-2y=2$ (11) $(13,7),(-11,-3)$

(12)(i) $y=12.5x-150$ (ii) 12seconds (iii) 80seconds

(13)(ii) $x-2y+4=0$ (iii) $2cm$ (iv) $14kg$ (v) $5cm$

(14)(i) $y=-\frac{71}{120}x+14.2, \quad 0\le x\le 24$

(ii) $y=f(\mathrm{x})$ is a periodic function with period $24, f(\mathrm{x})=f(x+24)$

(15)(i) The minimum length $=3280$units

(ii) $180, 360$ and 540units

(iii) The slope at each turning point is $\frac{9}{40}$

Exercise 6.3

(2) $5x-4y-15=0$ (3)(i) $\frac{8}{5}$ (ii) $\frac{23}{5}$

(4)(i) $x+3y+2=0$ (ii) $4x-3y-7=0$ (5) $x+5y-31=0$

(6)(i) $x+1=0$ (ii) $x-y=0$ (iii) $2x+y+3=0$

(7) $12x+5y+6=0$, and $12x+5y-20=0$ (8) $4x-3y+15=0$, and $4x-3y-25=0$

(9) $2x+3y-18=0$ (10) $7\sqrt{2}$, and $(-3,5)$

(12) (i)$\frac{14}{13}$, (ii)$\frac{5}{2}$ (13)(i) $4x-3y+k=0, k\in\mathbb{R}$ (ii) $3x+4y+k_1=0, k_1\in\mathbb{R}$

(14) $\sqrt{3}x-y-2\sqrt{3}=0$ (15) $A\left(\frac{13}{5},0\right)$ (16) $x+5y=\pm 10$

(17) $(0,7)$ (18)(i) $y=\begin{cases}1.50x, & 0\le x\le 10\\ x+5, & x>10\end{cases}$ (ii) ₹45

(19) $y=5x-7, \quad y=5x+10$

(20) $y+3=0,$ $2x+y+3=0,$ and $2x-y-3=0$